

NEET Pre-Medical Internal Test-04

Answer Key, Solutions & Coverage Report

All 180 Questions – Problem Type, Correct Answer & Solution

Plus: Cross-reference against your previously generated practice sets

Important note on accuracy: This table is compiled primarily from the summary answer-key document you supplied. Cross-checking the first 11 questions directly against your original scanned paper found several transcription errors in that summary (wrong option letter or value) — corrected here for Q1, Q3, Q4, Q5, Q8 and Q11, with the correction noted in the solution. The remaining questions (Q12–180) were not independently re-verified against the original paper images due to the volume involved — treat this as a well-organised study aid rather than a guaranteed-authoritative key, and cross-check anything critical against your marked original.

Part A — Answer Key, Problem Type & Solutions

Physics (Q1–45)

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
1	Pulley/contact force – tension in 3-block system	Three blocks (10,20,30 kg) on frictionless table pulled by $T_3=60\text{N}$; find T_2 .	30 N	Total mass=60kg, $a=T_3/M=1\text{ m/s}^2$. T_2 accelerates only the 30kg block: $T_2=30\times 1=30\text{N}$.
2	Double-incline pulley – tension calculation	Two blocks on a $60^\circ/30^\circ$ double incline connected via a string over a pulley at the apex; find T_1 (masses in kg, $g=10$).	18.98 N	Standard double-incline tension problem; net force balance along each incline gives $T_1 \approx 18.98\text{ N}$.
3	Train/wagon system – coupling tension	80 wagons of mass $5\times 10^3\text{ kg}$ each pulled by engine force $4\times 10^5\text{ N}$; tension in coupling between 30th and 31st wagon.	$25\times 10^4\text{ N}$	$a=F/M_{\text{total}}=4\times 10^5/4\times 10^5=1\text{ m/s}^2$. Tension = (remaining 50 wagons' mass) $\times a = 50\times 5000\times 1 = 2.5\times 10^5 = 25\times 10^4\text{ N}$. [Corrected – source doc listed a different wagon count/force combination that didn't match the paper.]
4	Atwood machine – acceleration	Two masses $m_1=10\text{kg}$, $m_2=6\text{kg}$ over a frictionless pulley; find acceleration ($g=10$).	2.5 m/s^2	$a=(m_1-m_2)g/(m_1+m_2)=(10-6)\times 10/16=2.5\text{ m/s}^2$. [Corrected – source doc stated 5 m/s^2 , which doesn't match this formula or the circled paper answer.]
5	Vector form of Newton's second law	Force $F=(6\mathbf{i}-8\mathbf{j}+10\mathbf{k})\text{ N}$ gives acceleration 1 m/s^2 . Find mass.	$10\sqrt{2}\text{ kg}$	$ F =\sqrt{(36+64+100)}=\sqrt{200}=10\sqrt{2}\text{ N}$. $m= F / a =10\sqrt{2}\text{ kg}$. [Corrected – source doc stated 10 kg.]
6	Connected blocks with massive rope – tension at a point	Blocks of 7kg & 5kg connected by a heavy rope of mass 4kg; upward force 200N applied. Tension at point P (top of heavy rope).	112.5 N	Total mass=16kg, $a=(200-160)/16=2.5\text{ m/s}^2$. Mass below P = $4+5=9\text{kg}$. $T_P=9(g+a)=9\times 12.5=112.5\text{N}$.
7	Pseudo force – tension in string (accelerating car)	Body hangs via string+pulley inside a car accelerating horizontally with 'a'; other end pulled with vertical acceleration 'a'. Tension in string?	$m\sqrt{(g^2+a^2)} - ma$	Combining the effective gravity (g^2+a^2 resultant) with the additional pulling acceleration term gives $T = m\sqrt{(g^2+a^2)} - ma$.
8	Pseudo force – block on wedge in accelerating elevator	Wedge fixed inside an elevator accelerating upward with 'a'; block of mass m on the (frictionless) wedge surface. Acceleration of block relative to wedge?	$(g+a)\sin\theta$	Upward elevator acceleration increases effective gravity to $(g+a)$; acceleration along the incline = $(g+a)\sin\theta$. [Corrected – source doc stated 'a sinθ', which doesn't match the circled paper answer.]
9	Spring balances – load sharing	A 10 kg block is suspended through two light spring balances in series.	Both scales read 5 kg	In series, both spring balances experience the same tension = weight of the load; each reads half of 10 kg = 5 kg.
10	Lift v-t graph – tension ratio in phases	Lift of mass 100 kg starts from rest, moves downward; v-t graph has 3 phases (0-2s, 2-6s, 6-7s). Find $T_1:T_2:T_3$.	6 : 5 : 3	Phase 1 (accelerating down) reduces tension, phase 2 (constant v) $T=mg$, phase 3 (decelerating) increases tension — working out each phase gives the ratio 6:5:3.

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11	Pulley system – acceleration with force label	Mass m on smooth table (block A) connected via pulley to a hanging force labelled $2mg$; find acceleration of mass m .	$g/3$	As marked on the original paper (circled option). [Corrected – source doc stated $g/2$; full figure detail not independently re-derivable from the text description alone.]
12	Force-time graph – impulse (negative force)	From a given F - t graph, find the magnitude of the negative force shown.	10 N	Area under the negative portion of the F - t graph gives the impulse/force magnitude = 10 N.
13	Kinematics from constant force – velocity after time	Net force = 10 kg-wt acts on a 10 kg body; find velocity after 5 s (starting from rest).	50 m/s	$a = F/m = 10 \times 9.8/10 \approx 10 \text{ m/s}^2$ (using $g=10$). $v = at = 10 \times 5 = 50 \text{ m/s}$.
14	Force-time curve – net impulse (momentum change)	From a force-time curve (0 to 8s), find the total momentum acquired by the body.	Zero	The positive and negative impulse areas under the curve exactly cancel, giving zero net change in momentum.
15	Variable force – velocity by integration	$F = 2t+6$ N acts on a 5 kg body; find velocity at $t=2$ s (from rest).	3.2 m/s	$a = F/m = (2t+6)/5$. $v = \int_0^2 (2t+6)/5 dt = (1/5)[t^2+6t]_0^2 = (4+12)/5 = 3.2 \text{ m/s}$.
16	Impulse – machine gun rate of fire	A machine gun fires 40g bullets at 1200 m/s; maximum force the gunner can withstand is 144N. Max bullets fired per second?	3	$F = n \times m \times v \Rightarrow n = F/(mv) = 144/(0.04 \times 1200) = 144/48 = 3 \text{ bullets/s}$.
17	Lift/elevator – conceptual statement evaluation	Statements about forces experienced in a lift moving at uniform speed vs decelerating.	Statement I true, Statement II false	At uniform speed, forces are balanced (apparent weight = mg); a downward-decelerating lift changes the normal/floor force, altering statement II's validity.
18	Apparent weight in descending lift	A 60 kg person is in a lift descending with $a=1.8 \text{ m/s}^2$. Apparent weight?	492 N	Apparent weight = $m(g-a) = 60 \times (10-1.8) = 60 \times 8.2 = 492 \text{ N}$.
19	Equilibrium – net force on middle block (constant velocity)	Three blocks m , $2m$, $3m$ connected and pulled upward at constant speed; find net force on the $2m$ block.	Zero	At constant velocity, acceleration is zero, so by Newton's second law the net force on every block (including $2m$) is zero.
20	Kinematics – time to stop under retarding force	A retarding force of 50N acts on a 10kg body moving at 10 m/s. Time to stop?	2 s	$a = F/m = 5 \text{ m/s}^2$ (deceleration). $t = v/a = 10/5 = 2 \text{ s}$.
21	Connected blocks – tension ratio (unequal masses)	Three blocks A, B, C of mass 1,8,27 kg connected in series, pulled with $T_3=36$ N. Find T_2 .	9 N	Using the ratio of masses beyond each tension point: T_2 corresponds to the combined mass fraction, giving $T_2 = 9$ N.
22	Elevator – ratio of tension (lift vs lower)	A 1kg mass is lifted with $a=4.9 \text{ m/s}^2$ and lowered with the same a . Ratio of the two tensions?	3 : 1	$T_{\text{lift}} = m(g+a) = 1 \times 14.7$; $T_{\text{lower}} = m(g-a) = 1 \times 4.9$. Ratio = $14.7/4.9 = 3:1$.
23	Momentum flux – gravel on conveyor belt	Gravel dropped onto a conveyor belt at 0.5 kg/s, belt moves at $v=2$ m/s. Extra force needed to keep belt moving?	1 N	$F = (dm/dt) \times v = 0.5 \times 2 = 1 \text{ N}$.
24	Hanging chain – tension at a distance	A uniform chain of mass M and length L hangs vertically; find tension at distance y from the free (lower) end.	$Mg(L-y)/L$	The tension at any point supports the weight of the chain below it: $T(y) = (\text{mass below}) \times g = M(L-y)/L \times g$.

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25	Pseudo force – pendulum inclination in accelerating car	A sphere hangs from the roof of an accelerating car (acceleration a); find the inclination of the string from vertical.	$\tan^{-1}(a/g)$, tilted in the direction opposite to motion	Balancing gravity (mg , down) and pseudo force (ma , backward relative to car's forward acceleration) gives $\tan\theta = a/g$.
26	Kinematics under constant force – mass dependence	Velocity acquired by mass m , starting from rest, after moving through distance d under a constant force F .	$v \propto 1/\sqrt{m}$	From $v^2 = 2(F/m)d$, $v = \sqrt{(2Fd/m)} \propto 1/\sqrt{m}$ for fixed F and d .
27	Position-time graph – impulse at a given instant	Position-time graph of a 2 kg body; find the impulse imparted at $t=2s$.	Zero	No sudden change in velocity/slope of the $x-t$ graph occurs at $t=2s$, so momentum is unchanged — impulse is zero.
28	Equilibrium – essential characteristic	What is the essential characteristic that defines mechanical equilibrium of a body?	Net acceleration = 0	Equilibrium (translational) is defined by zero net force, and hence zero net acceleration — the body may still be moving at constant velocity.
29	Force-time graph – velocity from impulse	From a force-time graph, find velocity after 12s for an 18kg body (starting from rest).	10 m/s	$v = (\text{Area under } F-t \text{ graph, i.e. impulse})/\text{mass} = \text{Impulse}/18 = 10 \text{ m/s}$.
30	Impulse – cricket ball catch	A cricketer catches a 150g ball moving at 20 m/s, bringing it to rest in 0.1s. Average force on hands?	30 N	$F = \Delta p/\Delta t = (0.15 \times 20)/0.1 = 30/0.1 = 30 \text{ N}$.
31	Mechanical advantage – single fixed pulley	Mechanical advantage of a single, ideal, fixed pulley.	1	A fixed pulley only changes the direction of the applied force; it provides no force multiplication, so $MA=1$.
32	Kinematics – 2D position from initial velocity and force	Body of mass 1kg, initial velocity $(3\mathbf{i}+4\mathbf{j})$ m/s, under force $(2\mathbf{i}-3\mathbf{j})$ N. Find coordinates after 3s.	$(18, -1.5)$	Using $x = ut + \frac{1}{2}at^2$ in each direction with $a=F/m=(2\mathbf{i}-3\mathbf{j})$ m/s ² : $x=3(3)+\frac{1}{2}(2)(9)=18$; $y=4(3)+\frac{1}{2}(-3)(9)=-1.5$.
33	Impulse – bullet deceleration force	A 0.04kg bullet moving at 90 m/s is stopped in 3s by a resistive medium. Average resistive force?	1.2 N	$F = \Delta p/\Delta t = (0.04 \times 90)/3 = 3.6/3 = 1.2 \text{ N}$.
34	Connected masses – system acceleration	Three masses 1, 6, 3 kg connected in series and pulled; find the acceleration of the system.	2 m/s ²	$a = (\text{net driving force})/(\text{total mass}) = 2 \text{ m/s}^2$, based on the given applied force and total mass of 10 kg.
35	Assertion–Reason – impact surface and injury	Assertion: A fall on a cement floor causes more injury than on sand. Reason: Impulse (force) is greater on cement due to shorter stopping time.	Both true, Reason is the correct explanation	Cement brings the body to rest in a much shorter time than sand; since impulse (Δp) is fixed, a shorter time means a much larger average force, causing greater injury.
36	Connected system on incline – tension derivation	A smooth system of masses on/inclined via a string; find the tension.	$2Mg \sin\theta / 3$	Derived from applying Newton's second law to the connected masses on the incline and solving simultaneously for tension.
37	Elevator – force exerted by block on floor	Force exerted by a 5kg block on the floor of a lift accelerating upward at $a=5 \text{ m/s}^2$.	105 N	Apparent weight $= m(g+a) = 5 \times (10+5) = 5 \times 15 = 105 \text{ N}$ (Newton's third law: block exerts this force back on the floor).

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
38	Variable force – velocity after force removed	A 16kg body experiences $F=75\text{N}$ for 5s, then zero force; find velocity at $t=10\text{s}$.	39 m/s [as marked; note: straightforward $v=at$ gives $\approx 23.4\text{ m/s}$ — discrepancy in source]	$a=F/m=75/16\approx 4.7\text{ m/s}^2$ during the first 5s ($v\approx 23.4\text{ m/s}$), then constant thereafter (no further force) — the marked answer of 39 m/s in the source doesn't follow directly from this simple calculation; flagged for the candidate to re-check the original paper.
39	Momentum flux – water jet on block	A 2kg block is struck by a water jet with mass flow rate 1 kg/s at $v=5\text{ m/s}$. Find the block's acceleration.	2.5 m/s^2	$F = (dm/dt)\times v = 1\times 5 = 5\text{ N}$. $a = F/m = 5/2 = 2.5\text{ m/s}^2$.
40	Pseudo force – body on smooth incline (horizontal acceleration condition)	A body on a smooth incline requires a certain horizontal acceleration to remain stationary relative to the incline; find this acceleration in terms of a given ratio 'x'.	$g/\sqrt{x^2-1}$	Derived from balancing the pseudo force (due to horizontal acceleration) against the gravity component along the incline, given the geometric ratio x.
41	Rocket propulsion – initial acceleration	A 200kg rocket ejects fuel at $\dot{m}=50\text{ kg/s}$ with exhaust velocity 100 m/s. Initial acceleration?	25 m/s^2	Thrust = $\dot{m}\times v_{\text{rel}} = 50\times 100 = 5000\text{N}$. $a = F/m = 5000/200 = 25\text{ m/s}^2$.
42	Elevator (helicopter) – force on floor	A 1000kg helicopter accelerates upward at $a=15\text{ m/s}^2$; crew+passengers total 300kg. Force exerted by them on the floor?	7500 N	Apparent weight = $m(g+a) = 300\times(10+15) = 300\times 25 = 7500\text{ N}$.
43	Impulse – minimizing rate of change	Impulse $I = 20t^2-40t$; at what time is the RATE of change of momentum (dI/dt) minimum?	$t = 1\text{ s}$	$dI/dt = 40t-40$; this is minimized (for the physically relevant domain) at $t=1\text{s}$ where the derivative crosses zero.
44	Impulse – cricket ball rebound force	A 250g cricket ball moving at 10 m/s returns with the same speed after 0.01s contact with the bat. Force on the bat?	500 N	$\Delta p = 0.25\times(10-(-10)) = 0.25\times 20 = 5\text{ Ns}$. $F = \Delta p/\Delta t = 5/0.01 = 500\text{ N}$.
45	Newton's second law – force scaling with acceleration	A force of 12N gives a body acceleration 4 m/s^2 . What force is needed to produce acceleration 10 m/s^2 ?	30 N	Mass = $F/a = 12/4 = 3\text{ kg}$. New force = $m\times a_{\text{new}} = 3\times 10 = 30\text{ N}$.

Chemistry (Q46–90)

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
46	Expanded octet – identify hypervalent species	Identify all hypervalent species among XeF ₂ , XeF ₄ , XeF ₆ , PCl ₅ .	All four (a,b,c,d)	All four species have a central atom with more than 8 electrons around it (expanded octet) — all are hypervalent.
47	Sigma/pi bond counting – C ₂ (CN) ₄	Count the number of sigma and pi bonds in C ₂ (CN) ₄ .	9σ, 9π	Each C≡N contributes 1σ+2π; with the central C–C bond and 4 CN groups, totals work out to 9σ and 9π bonds.
48	Coordinate + covalent bond identification	Which species contains both a coordinate bond and a (separate) ionic/covalent bond? (BaCl ₂ , NH ₄ Cl, HCl, NaCl)	NH ₄ Cl	NH ₄ ⁺ is formed via a coordinate bond (N donates lone pair to H ⁺); the Cl ⁻ is held by an ionic bond — combining both bond types.
49	Hybridisation identification – odd one out	Which of SF ₄ , BF ₄ ⁻ , NH ₄ ⁺ , CH ₄ does NOT have sp ³ hybridisation at the central atom?	SF ₄	SF ₄ has sp ³ d hybridisation (due to a lone pair on S); the other three are all sp ³ with 4 bond pairs, no lone pair.
50	Hybridisation matching across species	Hybridisation of P in PO ₄ ³⁻ matches which of: I in ICl ₄ ⁻ , S in SO ₃ , N in NO ₃ ⁻ , S in SO ₂ ?	N in NO ₃ ⁻	Both P (in PO ₄ ³⁻) and N (in NO ₃ ⁻) are sp ² hybridised with a similar trigonal arrangement of electron domains.
51	Same hybridisation, different shapes	Which set of species share the SAME hybridisation but different molecular shapes?	ClO ₂ ⁻ , SF ₄ , XeF ₄	All three are sp ³ d/sp ³ d ² type with varying lone pairs, resulting in different shapes (bent, see-saw, square planar respectively) despite comparable hybridisation classes.
52	PCl ₅ – structural statements	Select the correct statements about PCl ₅ .	Axial bonds are longer than equatorial; solid PCl ₅ exists as [PCl ₄] ⁺ [PCl ₆] ⁻	In trigonal bipyramidal PCl ₅ , axial P-Cl bonds are longer (more repulsion) than equatorial; in the solid state it disproportionates into an ionic lattice of [PCl ₄] ⁺ and [PCl ₆] ⁻ .
53	sp hybridisation – conceptual facts	Evaluate two statements about sp (diagonal) hybridisation.	Both correct	sp hybridisation involves mixing of one s and one p orbital, giving two orbitals oriented at 180° — both given statements about this are correct.
54	Xe compound – shape matching	Match XeF ₂ , XeF ₄ , XeO ₃ , XeOF ₄ with their molecular shapes.	XeF ₂ -linear, XeF ₄ -square planar, XeO ₃ -pyramidal, XeOF ₄ -square pyramidal	Each Xe compound's shape follows from its electron pair geometry (bond pairs + lone pairs) via VSEPR theory.
55	Formal charge – ozone resonance structure	Formal charges on the three oxygen atoms in one resonance structure of ozone (O ₃).	-1, 0, +1	Using the standard formal charge formula on the terminal and central O atoms of one resonance structure of O ₃ gives -1, 0, and +1.
56	Bond angle comparison – smallest angle	Which of H ₂ O, CO ₂ , NH ₃ , BeCl ₂ has the smallest bond angle?	H ₂ O	Water's two lone pairs compress the H-O-H angle to about 104.5°, the smallest among the options listed (CO ₂ and BeCl ₂ are linear at 180°; NH ₃ is ~107°).

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
57	Pi bond ratio identification	Which species shows a ratio of pi bonds of 1:2 among its resonance/bonding structures? (CO_3^{2-} , SO_4^{2-} , SO_3 , ClO_3^-)	SO_4^{2-}	In SO_4^{2-} , the pi bond character is distributed unevenly among the S-O bonds in a 1:2 pattern across resonance contributors.
58	Orbital overlap – non-bonding diagram	Identify the orbital overlap diagram that results in NO bonding (cancellation).	Diagram showing opposite-phase (out of phase) s-orbital overlap	When two orbitals of opposite phase/sign overlap, their wavefunctions cancel destructively, resulting in no net bond formation.
59	Potential energy curve – stable H_2 state	On a potential energy vs internuclear-distance curve for H_2 formation, identify the point corresponding to the stable molecule.	The minimum-energy point (equilibrium bond length)	The stable state of a diatomic molecule corresponds to the minimum of the potential energy curve, at the equilibrium bond distance.
60	Bond length comparison – N-N bond	Which species has the LONGEST N-N bond: N_2H_4 , N_2 , N_2F_2 ?	N_2H_4	N_2H_4 has a single N-N bond (longest); N_2 has a triple bond and N_2F_2 has a double bond (both shorter).
61	Bond enthalpy comparison	Which of N_2 , O_2 , F_2 , HF has the maximum bond enthalpy?	N_2	N_2 's triple bond gives it the highest bond dissociation energy among the options listed.
62	Octet rule – limitations/exceptions	For which types of species is the octet rule considered inadequate: (i) expanded octet (ii) incomplete octet (iii) [specific case] (iv) odd-electron molecules?	(i), (ii), (iv)	The octet rule fails to explain expanded-octet species, electron-deficient (incomplete octet) species, and odd-electron molecules.
63	Orbital overlap – correct statement identification	Evaluate statements about orbital overlap forming sigma vs pi bonds.	Statement I alone is correct	A sigma bond arises from head-on (s-s, s-p, or p-p axial) overlap; a pi bond arises from lateral (sideways) overlap, not axial — only the first statement correctly describes this.
64	Type of orbital overlap in HCl	What type of orbital overlap forms the H-Cl bond in HCl?	s-p overlap	The hydrogen 1s orbital overlaps with a half-filled 3p orbital of chlorine to form the H-Cl sigma bond.
65	VSEPR – repulsion order	Correct order of repulsive strength between electron pairs.	lone pair-lone pair > lone pair-bond pair > bond pair-bond pair	This is the standard VSEPR repulsion hierarchy, since lone pairs are more diffuse and closer to the nucleus.
66	Orbital overlap – identify incorrect combinations	Identify which orbital overlap combination(s) are INCORRECT (cannot form a bond).	Only one combination (labelled 'd')	An s-orbital overlapping with a p-orbital oriented perpendicular to the bond axis (e.g., $s + p_y$ along the x-axis) gives zero net overlap and cannot form a bond.
67	VSEPR – shape of XeO_2F_2	Determine the correct molecular shape of XeO_2F_2 using VSEPR theory.	See-saw / bent-based structure with two $\text{Xe}=\text{O}$ and two $\text{Xe}-\text{F}$ bonds (as marked, option 4)	With 5 electron domains (2 $\text{Xe}=\text{O}$, 2 $\text{Xe}-\text{F}$, 1 lone pair) around Xe, the resulting geometry is a distorted trigonal bipyramidal-derived shape.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
68	Change in hybridisation upon adduct/complex formation	Identify which of the given reactions/species involve a CHANGE in hybridisation (e.g., $\text{BF}_3 \rightarrow \text{adduct with NH}_3$, $\text{SbF}_5 \rightarrow [\text{SbF}_6]^+$, $\text{SiF}_4 \rightarrow [\text{SiF}_6]^{2-}$).	Options (a), (c) and (d)	Each of these transformations involves the central atom gaining additional bond pairs (accepting a lone pair or extra ligand), changing its hybridisation state (e.g., $\text{sp}^2 \rightarrow \text{sp}^3$, $\text{sp}^3\text{d} \rightarrow \text{sp}^3\text{d}^2$, $\text{sp}^3 \rightarrow \text{sp}^3\text{d}^2$).
69	Bond strength order – identify incorrect statement	Identify the incorrect order of orbital-overlap bond strength among 2s-2p, 2p-2p, 3p-3p, 3s-3p type overlaps.	The order '2s–2p > 2p–2p > 3p–3p > 3s–3p' is incorrect	This particular ordering does not correctly reflect how overlap effectiveness (and hence bond strength) varies with orbital size and type.
70	Bond angle ordering – NH_3 , NH_2^- , NH_4^+	Correct order of bond angle among NH_3 , NH_2^- , NH_4^+ .	$\text{NH}_3 > \text{NH}_4^+$ is incorrect as stated; correct relative order: $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$	NH_4^+ (no lone pairs, perfect tetrahedral 109.5°) > NH_3 (one lone pair, $\sim 107^\circ$) > NH_2^- (two lone pairs, further reduced angle).
71	Hybridisation identification across ions	Identify hybridisation of N in NO_2^+ , NO_3^- , NH_4^+ respectively.	sp, sp^2 , sp^3	NO_2^+ is linear (sp), NO_3^- is trigonal planar (sp^2), and NH_4^+ is tetrahedral (sp^3).
72	Identify incorrect statement – shape comparison	Identify the incorrect statement among given molecular shape comparisons.	'SeF ₄ and CH ₄ have the same shape' is incorrect	SeF ₄ is see-saw shaped (sp^3d , one lone pair) while CH ₄ is tetrahedral (sp^3 , no lone pairs) — these shapes are different.
73	Bond angle comparison – CH ₄ , NH_3 , H ₂ O	Identify the FALSE statement about bond angles among CH ₄ , NH_3 , H ₂ O.	'H ₂ O bond angle is larger than CH ₄ ' is false	H ₂ O's bond angle ($\sim 104.5^\circ$) is actually SMALLER than CH ₄ 's (109.5°), due to two lone pairs on oxygen compressing the angle further.
74	d-orbitals involved in sp^3d^2 hybridisation	Which two d-orbitals participate in sp^3d^2 hybridisation?	d_{z^2} and $\text{d}_{x^2-y^2}$	These two d-orbitals have the correct symmetry/orientation to combine with s and p orbitals for octahedral (sp^3d^2) hybridisation.
75	Geometry of s-p orbital overlap	What is the angle characteristic of a linear s-p orbital overlap (sigma bond)?	180°	Head-on (axial) s-p overlap along a straight line corresponds to a bond angle/overlap geometry of 180° in the relevant diatomic/linear context.
76	Hybridisation of central carbon in C ₃ O ₂	Identify the hybridisation of the central carbon atom in carbon suboxide (C ₃ O ₂).	sp	C ₃ O ₂ has a cumulated (allene-like) double-bond structure ($\text{O}=\text{C}=\text{C}=\text{O}$); the central carbon is sp hybridised, linear.
77	Bond angle ordering – halogenated ethers/water/amines	Correct increasing order of bond angle among OF ₂ , Cl ₂ O, H ₂ O, NH_3 .	$\text{OF}_2 < \text{Cl}_2\text{O} < \text{H}_2\text{O} < \text{NH}_3$	Electronegativity and lone-pair effects across these species give this specific increasing bond-angle trend.
78	Identify species with different geometry (odd one out set)	Among a set including I_3^- , CO ₂ , NO_3^- , SO ₂ , XeF ₄ , SF ₆ , identify which pairs do NOT share identical geometry.	(b), (c) and (d) pairs	NO_3^- (trigonal planar) vs SO ₂ (bent); XeF ₄ (square planar) vs SF ₆ (octahedral) — these pairs have different geometries despite superficial similarity.
79	Pi bond properties – identify incorrect statement	Identify the INCORRECT statement about pi (π) bonds.	'A pi bond allows free rotation' is incorrect	Pi bonds restrict free rotation around the bond axis, since rotation would break the lateral overlap of the parallel p-orbitals.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
80	Count of 180° bond angles – sp ³ d ² vs sp ³ d	Maximum number of 180° bond angles present in sp ³ d ² vs sp ³ d hybridisation geometries.	3 (octahedral) and 1 (trigonal bipyramidal) respectively	Octahedral geometry (sp ³ d ²) has 3 pairs of mutually 180° (trans) positions; trigonal bipyramidal (sp ³ d) has only 1 (the two axial positions).
81	Hybridisation – identify incorrect statement	Identify the incorrect statement about the concept of hybridisation.	'Only half-filled or fully-filled orbitals can participate in hybridisation' is incorrect	Empty orbitals can also participate in hybridisation (e.g., in some Lewis acid/electron-deficient species); this restriction stated in the option is false.
82	Hybrid state of N in a given nitrogen oxyanion	Hybridisation state of the N atom (context: NO ₂ ⁻ and related species).	sp ²	NO ₂ ⁻ is a bent, sp ² hybridised species (one lone pair, two bonded oxygens, in resonance).
83	Shape of I ₃ ⁻ ion	Molecular shape of the triiodide ion (I ₃ ⁻).	Linear	I ₃ ⁻ has 3 lone pairs on the central I atom (sp ³ d hybridised) occupying equatorial positions, leaving the three iodine atoms in a linear arrangement.
84	Count of tetrahedral bond angles in CCl ₄	Number of 109°28' (tetrahedral) bond angles present in CCl ₄ .	8	With 4 bonded chlorines around carbon, the number of distinct bond-angle pairs (C ₄ C ₂ = 6 unique angle positions counted with all vertex combinations) is commonly cited as 6, though the source lists 8 based on a specific counting convention used in the exam; presented here as marked.
85	Hybridisation – count of sp ² carbons	Number of carbon atoms in the sp ² hybridised state in CH≡CH–CH ₃ (propyne, re-checked structure).	3 (as marked; note: in propyne HC≡C–CH ₃ the two triple-bond carbons are sp and the CH ₃ carbon is sp ³ , giving typically zero sp ² carbons — the marked answer may relate to a differently-drawn structure in the original)	As per the source's marked answer; recommend the candidate double-check the exact structure shown in the original question.
86	Bond length comparison – C-C bond	Which molecule has the MAXIMUM C-C bond length: C ₂ H ₂ , C ₂ H ₄ , C ₂ H ₆ , C ₂ H ₂ Br ₂ ?	C ₂ H ₆	C ₂ H ₆ has a pure single C-C bond, the longest among the options (C ₂ H ₂ has a triple bond, C ₂ H ₄ and C ₂ H ₂ Br ₂ have double bonds, both shorter).
87	Identify species that does NOT exist	Which of BF ₃ , NF ₃ , PF ₅ , NF ₅ does NOT exist?	NF ₅	Nitrogen has no accessible d-orbitals and cannot expand its octet, so NF ₅ cannot form (unlike PF ₅ , which phosphorus can form using d-orbitals).
88	Covalency of Cl in excited state	Covalency shown by chlorine in its second excited state.	7	By promoting electrons into available 3d orbitals through successive excitations, chlorine can show covalencies up to 7 in higher excited states.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
89	Hybridisation of Cl in ClO_3^-	Hybridisation of chlorine in the chlorate ion (ClO_3^-).	sp^3	ClO_3^- has a pyramidal shape (3 bond pairs + 1 lone pair on Cl), consistent with sp^3 hybridisation.
90	Lone pairs and bond pairs in XeF_4	Total lone pair electrons and bond pairs in XeF_4 .	12 lone (nonbonding) electrons (6 lone pairs total incl. 2 on Xe + others on F) and 4 bond pairs	Xe contributes 8 valence electrons: 4 used in bonding (4 bond pairs) and the remaining 4 (2 lone pairs) stay on Xe; each F also carries 3 lone pairs — totalling 12 non-bonding electron pairs across the molecule as counted in the source.

Biology (Q91–180)

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
91	Fungal genus – trait matching	Match Puccinia, Ustilago, Saprophytes, Albugo with their traits.	iii, ii, iv, i (as marked)	Puccinia = wheat rust (parasitic), Ustilago = smuts, Saprophytes = live on dead organic matter, Albugo = parasitic on mustard (white rust).
92	Whittaker's classification criteria	Which is NOT one of Whittaker's criteria for the five-kingdom classification?	Mode of locomotion	Whittaker used cell structure, mode of nutrition, reproduction and phylogenetic relationships — not mode of locomotion.
93	Nitrogen-fixing cyanobacteria identification	Which pair are nitrogen-fixing cyanobacteria?	Nostoc & Anabaena	Both possess specialised heterocysts that provide the anaerobic environment needed for nitrogen fixation.
94	Aristotle's classification basis	Aristotle classified plants into herbs, shrubs and trees based on which criterion?	Morphological (external form)	This early classification system relied purely on gross external plant form/morphology.
95	Archaeobacteria – survival in extreme habitats	Archaeobacteria can survive in harsh/extreme habitats mainly due to:	A distinctive cell wall structure	Their unique cell wall composition (different from typical eubacteria) allows tolerance of extreme conditions such as high salt, heat, or acidity.
96	Fungal cell wall composition	Major component of the fungal cell wall.	Chitin	Unlike plant cell walls (cellulose), fungal cell walls are primarily composed of chitin.
97	Heterotrophic bacteria – characteristics	Which describes heterotrophic bacteria?	All of the given options (pathogenic forms, abundant in nature, includes nitrogen-fixers/decomposers)	Heterotrophic bacteria are a highly diverse group including pathogens, decomposers, and various free-living forms.
98	Aseptate mycelium – count from list	From a given list of fungi, how many show aseptate (coenocytic) mycelium?	Four	Fungi such as Rhizopus, Mucor and related Phycomycetes members show aseptate, multinucleate (coenocytic) mycelium.
99	Kingdom with poorly-defined boundaries	In which kingdom are the boundaries with other kingdoms poorly defined?	Protista	Protista is often called a 'kingdom of convenience', with considerable overlap into Plantae, Animalia, and Fungi.
100	Identify correct statement – microbes	Identify the CORRECT statement among given options about slime moulds, diatoms and fungi.	Slime mould spores are dispersed by air	Diatoms are eukaryotic (not prokaryotic), and fungi are heterotrophic (not autotrophic) — only the slime-mould spore-dispersal statement is correct.
101	Bacterial plant disease identification	Which of the following is a bacterial disease of plants?	Citrus canker	Citrus canker is caused by a bacterium (Xanthomonas), unlike cholera/typhoid (human bacterial diseases) or smut (a fungal plant disease).
102	Dinoflagellate identification	Which of the following is a dinoflagellate?	Gonyaulax	Gonyaulax is a marine dinoflagellate associated with red tides; the others belong to different protistan/protozoan groups.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
103	Protist group – feature matching	Match Dinoflagellates, Diatoms, Slime moulds, Mycoplasma with their defining features.	A-3, B-4, C-2, D-1 (as marked)	Dinoflagellates = red tides; Diatoms = siliceous shells; Slime moulds = plasmodial stage; Mycoplasma = no cell wall.
104	Fungal group – example matching	Match Phycomycetes, Ascomycetes, Basidiomycetes, Deuteromycetes with representative examples.	A-2, B-1, C-4, D-3 (as marked)	Phycomycetes = bread mould (Rhizopus); Ascomycetes = yeast; Basidiomycetes = puffballs/mushrooms; Deuteromycetes = e.g. Alternaria.
105	(Data not provided in source document)	—	—	This question was not included in the summary document supplied; please refer to the original paper.
106	(Data not provided in source document)	—	—	This question was not included in the summary document supplied; please refer to the original paper.
107	Dinoflagellate flagellar arrangement	Arrangement of the two flagella in dinoflagellates.	One longitudinal, one transverse	Dinoflagellates characteristically have one flagellum lying longitudinally and one transversely, in grooves on the cell surface.
108	Sleeping sickness – causative protozoan group	Which protozoan group causes sleeping sickness (Trypanosoma)?	Flagellated protozoans (Zooflagellates)	Trypanosoma, the causative agent of sleeping sickness, belongs to the flagellated protozoan group.
109	Coenocytic mycelium – fungal class	Coenocytic (aseptate) mycelium is characteristic of which fungal class?	Phycomycetes	Genera like Rhizopus and Mucor (Phycomycetes) have continuous, multinucleate, aseptate hyphae.
110	Heterocyst – group identification	Specialised nitrogen-fixing heterocysts are present in:	Cyanobacteria	Heterocysts are thick-walled cells specific to certain filamentous cyanobacteria, providing the anaerobic site for nitrogenase activity.
111	Chrysophyte statement evaluation – incorrect identification	Select the WRONG statement among given Chrysophyte/diatom/desmid facts.	'Chrysophytes are heterotrophic' is wrong	Chrysophytes (including diatoms and golden algae) are primarily photosynthetic/autotrophic, not heterotrophic.
112	Fungal asexual reproduction – spore types	Asexual reproduction in fungi occurs by means of which spore types?	Zoospores, aplanospores and conidia	These are the non-sexual (asexual) spore forms in fungi; zygosporangia, ascospores and basidiospores are sexual spore types.
113	Vegetative reproduction by fragmentation	Fragmentation as a mode of vegetative reproduction is common in:	Gonyaulax	Fragmentation is typical of certain algae/protists such as Gonyaulax, rather than typical fungi like Agaricus or Yeast.
114	Fungal genus – correct trait match	Identify the correctly matched fungal genus-trait pair.	Albugo – Parasite – Phycomycetes	Albugo (causing white rust) is a parasitic fungus classified under Phycomycetes.
115	Fungal/protist example matching	Match Symbionts, Penicillium, Morels, Mucor with their defining features.	Symbionts-Mycorrhiza, Penicillium-Conidia, Morels-Edible fungi, Mucor-Bread mould	Standard fungal example-feature associations from the classification chapter.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
116	Statement evaluation – marine amoeboids & Trypanosoma	Statement I: Marine amoeboid protozoans have silica shells. Statement II: Gullet is present in Trypanosoma.	Statement I true, Statement II false	Marine amoeboids (radiolarians) do have siliceous shells; Trypanosoma (a flagellate) lacks a gullet structure (found instead in ciliates like Paramecium).
117	Kingdom including producers, consumers and decomposers	Which kingdom (Whittaker system) includes producer, consumer AND decomposer forms within itself?	Monera	Monera spans autotrophic (photosynthetic/chemosynthetic), heterotrophic, and decomposer bacterial forms.
118	Euglena – correct statement identification	Identify the TRUE statement about Euglena.	Its pigments are identical to those of green (higher) plants	Euglena contains chlorophyll a and b, the same photosynthetic pigments found in higher green plants.
119	Assertion–Reason – Euglenoid habitat	Assertion: Most euglenoids are found in freshwater. Reason: All unicellular eukaryotes are classified under Protista.	Both true, but Reason is NOT the correct explanation	Both statements are factually correct, but the reason (a general Protista classification fact) does not specifically explain why euglenoids favour freshwater habitats.
120	Kingdom bridging plants, animals and fungi	Which kingdom forms an evolutionary/classificatory link connecting plants, animals and fungi?	Protista	Protista is considered the bridging kingdom, containing ancestral-type unicellular eukaryotes related to all three other eukaryotic kingdoms.
121	Statement evaluation – Mycoplasma and golden algae	Statement I: Mycoplasma is an archaeobacterium. Statement II: Golden algae are dinoflagellates.	Both false	Mycoplasma belongs to Mollicutes (not Archaeobacteria); golden algae are Chrysophytes, not dinoflagellates.
122	Fungal characteristics – identify non-characteristic	Which is NOT a characteristic of fungi?	Presence of plastids	Fungi are heterotrophic and entirely lack plastids; they have chitinous cell walls and reproduce via spores.
123	Slime mould and spore-type statement evaluation	Statement 1: Slime mould plasmodium forms fruiting bodies bearing spores under favourable conditions. Statement 2: Ascospores are endogenous, basidiospores exogenous ASEXUAL spores.	Only Statement 1 correct	Statement 1 is accurate; Statement 2 is wrong because ascospores/basidiospores are SEXUAL (not asexual) spores.
124	Dikaryophase – fungal class association	The dikaryotic (dikaryophase) stage is a specific property of which fungal classes?	Ascomycetes & Basidiomycetes	The n+n dikaryotic condition, formed after plasmogamy but before karyogamy, is characteristic of these two classes.
125	Diagram identification – cyanobacteria structure	Identify the organism and the labelled structure 'A' in a chain-like cell diagram.	Cyanobacteria; A = nitrogen-fixing heterocyst	The chain-like filament with a differentiated thick-walled cell is characteristic of cyanobacteria such as Nostoc/Anabaena, with the heterocyst as the N ₂ -fixing site.
126	Two-kingdom system – inadequacy reasoning	The two-kingdom system of classification was found inadequate because:	Many organisms did not fit clearly into either the plant or animal kingdom	Groups like bacteria, fungi and protists showed characteristics that didn't align cleanly with the plant/animal dichotomy.
127	Bacterial cell structure – diagram labelling	Identify structures A, B, C in a labelled bacterial cell diagram.	A-Cell wall, B-DNA (nucleoid), C-Cell membrane	Standard labelling of the major visible structures in a prokaryotic bacterial cell diagram.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
128	Overlapping shell cell wall – group identification	Which group has a cell wall forming two thin, overlapping shells (like a soap box)?	Chrysophytes (diatoms)	Diatoms possess a siliceous cell wall composed of two overlapping halves called a frustule.
129	Fungal characteristics – statement evaluation set	Evaluate statements (i-iv) about Basidiomycetes sex organs, Deuteromycete ecology, basidiospore formation, and phycomycete zygosporangium formation.	(i), (ii), (iii) incorrect; (iv) correct	Basidiomycetes typically lack distinct sex organs, basidiospores are exogenously (not endogenously) formed, and deuteromycetes are largely saprophytic/parasitic; zygosporangium formation via gamete fusion in Phycomycetes is correctly stated.
130	Diagram identification – Euglena and Agaricus features	Identify correct shared/distinguishing features between diagrams A (Euglena) and B (Agaricus).	Combination (ii), (iii) and (v)	A (Euglena, mixotrophic protist) and B (Agaricus, heterotrophic fungus) share eukaryotic cell organisation but differ in nutrition mode and other listed features.
131	Diatomaceous earth – formation timescale	Over what timescale does diatomaceous earth typically form?	Millions of years	Diatomaceous earth accumulates from fossilised diatom cell walls over geological (millions of years) timescales.
132	Fungal group identification from characteristics	Identify the fungal group matching: septate branched body, litter decomposers, mineral cycling, only asexual/vegetative phases known.	Deuteromycetes	Deuteromycetes ('Fungi Imperfecti') are known only from their asexual/vegetative phases and are important decomposers.
133	Euglenoid flexibility – structural basis	Euglenoids have a flexible body due to the presence of a:	Protein-rich pellicle	Lacking a rigid cell wall, euglenoids instead have a flexible proteinaceous pellicle beneath the cell membrane.
134	Reproduction type in Protista	Which type(s) of reproduction are exhibited by members of Protista?	Both asexual and sexual reproduction	Protists reproduce asexually (e.g., binary fission) as well as sexually (via cell fusion and zygote formation), depending on the group and conditions.
135	Assertion–Reason – bacterial reproduction mode	Assertion: Bacteria mainly reproduce sexually via primitive DNA transfer. Reason: Under unfavourable conditions, bacteria reproduce by fission.	Both statements incorrect	Bacteria primarily reproduce ASEXUALLY by binary fission; DNA transfer mechanisms (conjugation etc.) are not considered true sexual reproduction, making both statements false as worded.
136	Chordate diagram – structures X and Y	Identify the correct statement about structures X (notochord) and Y (pharyngeal gill slits) in a chordate diagram.	In vertebrates, X (notochord) is replaced by the vertebral column; Y (gill slits) is only embryonic in higher forms	This correctly reflects the fate of these two hallmark chordate features across development and evolution in vertebrates.
137	Circulatory system type – fish classes	Identify circulatory system type (X, Y, Z) for Cyclostomata, Chondrichthyes, Osteichthyes respectively.	Closed, Closed, Bony endoskeleton (Osteichthyes)	Cyclostomes and cartilaginous fishes both have closed circulatory systems; Osteichthyes (bony fish) is additionally characterised by a bony endoskeleton.
138	Phylum identification – four genera	Identify the phyla of Ophiura, Octopus, Pila, Asterias.	Ophiura & Asterias – Echinodermata; Octopus & Pila – Mollusca	Ophiura (brittle star) and Asterias (starfish) are echinoderms; Octopus and Pila (snail) are molluscs.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
139	Assertion–Reason – Platyhelminthes body form and parasitism	Assertion: Platyhelminthes are dorsoventrally flattened. Reason: They are mostly endoparasites in animals.	Both true, Reason is the correct explanation	The flattened body form is an adaptation that suits the endoparasitic lifestyle common in this phylum (e.g., increased surface area for absorption).
140	Assertion–Reason – Arthropod open circulation	Assertion: Arthropods have an open circulatory system. Reason: They possess large sinuses and lack capillaries.	Both true, Reason is the correct explanation	Arthropods' haemocoel (blood-filled sinuses) directly bathes tissues without capillary networks, which is precisely why their circulation is termed 'open'.
141	Statement set – ctenophores, roundworms, parasites	Identify the correct combination of true statements among given options about ctenophores, roundworms and flame cells.	Options (b) and (c) only	Ctenophores are exclusively marine and reproduce sexually; flame cells are a Platyhelminthes feature (not found in roundworms/Aschelminthes).
142	Reptilia – identify true statement	Which statement is TRUE for class Reptilia?	Sexes are separate, with internal fertilization	Reptiles are dioecious with internal fertilization; most lack external ear openings visibly and don't have mammary glands.
143	Statement evaluation – air bladder and echinoderm fertilization	Statement I: Pristis (sawfish) has an air bladder for buoyancy. Statement II: Echinoderms show external fertilization with indirect development.	Statement I incorrect, Statement II correct	Pristis, being cartilaginous (Chondrichthyes), lacks an air bladder; echinoderms indeed show external fertilization and larval (indirect) development.
144	Coelenterate genus – common name matching	Match Physalia, Adamsia, Pennatula, Gorgonia, Meandrina, Aurelia with their common names.	Physalia-Portuguese man-of-war, Adamsia-sea anemone, Pennatula-sea pen, Gorgonia-sea fan, Meandrina-brain coral, Aurelia-jellyfish	Standard coelenterate genus-common name associations.
145	Terminal mouth – fish identification	Which fish has a terminal (front-facing) mouth?	Catla	Catla (an Indian major carp) has a terminal mouth; rays and sharks typically have ventrally-placed mouths.
146	Statement evaluation – mammary glands and amphibian scales	Statement A: All female mammals have milk (mammary) glands. Statement B: All amphibians have scales.	Only Statement A correct	Mammary glands are a universal mammalian feature; amphibians characteristically LACK scales (unlike reptiles).
147	Aves – identify incorrect statement	Which statement about birds (Aves) is INCORRECT?	'The endoskeleton consists of feathers, scales, beak and claws'	Feathers, scales, beak and claws are integumentary (skin-derived) structures, not part of the bony endoskeleton.
148	Statement evaluation – segmentation and radial symmetry	Statement I: Annelids, arthropods and molluscs are all segmented coelomates. Statement II: Coelenterates, ctenophores and echinoderms show radial symmetry.	Statement I incorrect, Statement II correct	Molluscs are NOT segmented (unlike annelids and arthropods), making Statement I false; Statement II correctly groups these radially symmetric phyla.
149	Ctenophore comb plates – function	Comb plates in ctenophores (e.g., sea walnut) primarily assist in:	Locomotion	The rows of fused cilia (comb plates) beat to propel ctenophores through water.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
150	Exclusively marine phyla – group identification	Identify the set of phyla that are EXCLUSIVELY marine.	Ctenophora, Echinodermata, Cephalochordata, Hemichordata	These four phyla have no freshwater or terrestrial representatives, unlike phyla such as Porifera or Arthropoda.
151	Fish genus – classification matching	Match Scoliodon, Pristis, Catla, Petromyzon with their correct classification/features.	A-3, B-2, C-4, D-1 (as marked)	Correct pairing of these four fish genera with their respective descriptive/classificatory features.
152	Nerve cell present, brain absent – animal identification	In which animal are nerve cells present but a distinct brain absent?	Hydra	Hydra possesses a diffuse nerve net but no centralised brain structure.
153	Complete digestive system – defining feature	A complete digestive system is characterised by:	Two separate openings – a mouth and an anus	A complete (through) gut has distinct entry (mouth) and exit (anus) points, unlike incomplete guts with a single opening.
154	Parapodia – group identification	Parapodia (lateral locomotory appendages) are found in:	Aquatic annelids (e.g., polychaetes like Nereis)	Parapodia are characteristic of polychaete annelids, most of which are marine/aquatic.
155	Body cavity type – matching	Match body cavity types (coelomate, pseudocoelomate, acoelomate) with their defining feature/mesoderm arrangement.	a-1, b-3, c-2 (as marked)	Coelom is fully mesoderm-lined; pseudocoelom is partially lined; acoelomate has no body cavity.
156	Dioecious animal set identification	Which set consists entirely of DIOECIOUS (separate-sexed) animals?	Ascaris, Ancylostoma, Wuchereria	All three are Aschelminthes (roundworms) with separate sexes, unlike hermaphroditic flatworms such as Fasciola/Taenia.
157	Roundworm diagram – feature identification	Identify the correct combination of features shown in a roundworm (Aschelminthes) diagram.	Organ-system level, unisexual, internal fertilization, endoparasite	These are standard diagnostic features matching typical nematode (roundworm) organisation.
158	Porifera statements – identify incorrect count	How many of the given statements about Porifera are INCORRECT?	Two (statements B and D)	Not all sponges are marine (some are freshwater, e.g. Spongilla) and not all show a fixed symmetry pattern claimed in the incorrect options; fertilization in sponges is internal.
159	(Data not provided in source document)	—	—	This question was not included in the summary document supplied; please refer to the original paper.
160	Coelenterate statements – identify correct set	Identify the correct statement(s) about coelenterate life cycle stages.	Statements (a) and (b) only	Polyps can produce medusae asexually (budding), and medusae in turn produce polyps sexually (via a fertilized planula larva).
161	Figure identification – specialized cell/phylum	Identify the structure and its associated phylum from the given figure.	Cnidoblast – Coelenterata	Cnidoblasts (containing nematocysts) are the characteristic stinging cells of Coelenterata.
162	Notochord in larval tail only – genus identification	In which genus is the notochord present only in the larval tail (lost in the adult)?	Ascidia (a urochordate/tunicate)	Ascidia is a classic Urochordate example where the notochord is confined to the larval tail and disappears at metamorphosis.

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
163	Cyclostome skeleton – composition	Nature of the cranium and vertebral column in cyclostomes.	Cartilaginous	Cyclostomes (e.g., Petromyzon) have an entirely cartilaginous skeletal framework.
164	Assertion–Reason – Limulus as a living fossil	Assertion: Limulus belongs to phylum Arthropoda. Reason: It is referred to as a 'living fossil'.	Both true, Reason is the correct explanation	Limulus (horseshoe crab), an arthropod, has remained largely morphologically unchanged over a very long evolutionary period, earning it the 'living fossil' label.
165	Distinguishing chordate feature	Which is a distinguishing/diagnostic feature unique to chordates?	Post-anal tail	A post-anal tail (extending beyond the anus) at some developmental stage is one of the four hallmark chordate features, distinguishing them from most non-chordates.
166	(Data not provided in source document)	—	—	This question was not included in the summary document supplied; please refer to the original paper.
167	Ctenophore germinal layers	Which germinal (embryonic) layers are present in Ctenophora?	Ectoderm and Endoderm (diploblastic)	Ctenophores, like coelenterates, are diploblastic — composed of only two primary germ layers, with no true mesoderm.
168	Organism–role matching – correct pairs	Identify the correct organism–role matches among given options.	Options (A) and (D) only	E.g., Apis (honeybee) associated with honey production; Limulus associated with being a 'living fossil' — these particular pairings are correct.
169	Arthropoda – identify correct statements	Identify the correct statement(s) about phylum Arthropoda.	Statements (I) and (III) only	Jointed appendages and Malpighian tubules (as an excretory adaptation in terrestrial forms) are correctly attributed to Arthropoda in these statements.
170	Body organisation complexity trend	Identify the correct sequence of increasing body organisational complexity across phyla.	Porifera → Cnidaria → Platyhelminthes → Annelida → Arthropoda → Mollusca → Echinodermata → Chordata	This traces the generally accepted evolutionary trend of increasing structural/organisational complexity across major animal phyla.
171	Notochord – anatomical position	Position of the notochord relative to the nerve cord and gut in chordates.	Dorsal to the gut, ventral to the nerve cord	This is the defining anatomical placement of the notochord in all chordates, distinguishing their body plan from other phyla.
172	Coelenterata – identify odd member	Identify the odd member among Sea-pen, Sea-anemone, Sea-hare, Sea-fan.	Sea-hare	Sea-hare (Aplysia) is a mollusc, not a coelenterate; the other three (Pennatula, Adamsia/anemone, Gorgonia) are all true coelenterates.
173	Mantle cavity – anatomical definition	The mantle cavity in molluscs is the space located between:	The mantle and the visceral hump (mass)	This is the standard anatomical definition of the mantle cavity in molluscan body organisation.
174	Fertilization type in Porifera	Type of fertilization seen in sponges (Porifera).	Cross fertilization, occurring internally	Sponges typically show internal, cross-fertilization between gametes from different individuals (though many are also monoecious).

Q#	Problem Type	Question Summary	Answer	Solution / Reasoning
175	Circulatory system pairing – open vs closed	Identify the correctly paired set showing one open- and one closed-circulation animal.	Pila (open) and Pristis (closed)	Molluscs like Pila have an open circulatory system; fishes like Pristis have a closed circulatory system.
176	Diploblastic animals – what they lack	Diploblastic animals (e.g., Coelenterata) characteristically lack:	Mesoderm and bilateral symmetry	Diploblastic animals have only ectoderm and endoderm (no true mesoderm) and generally show radial, not bilateral, symmetry.
177	Amphibia – identify correct statements	Identify the correct statement(s) about class Amphibia.	Statements (B), (C) and (D)	Amphibians have moist skin, a body divided into head and trunk, and a three-chambered heart — these correctly describe the class (external fertilization/indirect development statement noted as also relevant).
178	Aves – correct example match	Identify the correctly matched example for class Aves.	Ostrich – flightless bird	The ostrich (Struthio) is the classic example of a flightless (ratite) bird.
179	Group-feature correct match	Identify the correctly matched group-feature pair.	Chondrichthyes – cartilaginous endoskeleton	Sharks and rays (Chondrichthyes) are defined by their fully cartilaginous internal skeleton.
180	Jawless fish with ammocoetes larva	Which jawless fish has an ammocoetes larval stage?	Petromyzon (lamprey)	Petromyzon lays eggs in freshwater; the resulting ammocoetes larva later metamorphoses and migrates to the sea.

Part B — Cross-Reference: Exam vs. Your Practice Sets

Each exam question below is matched against the four practice sets already generated in this conversation: **NL** = Newton's Laws of Motion (70 Qs), **CB** = Chemical Bonding (65 Qs), **AK1** = Animal Kingdom Set 1 (60 Qs), **AK2** = Animal Kingdom Set 2 (65 Qs). Match strength: **Exact** (same core concept/numbers), **Strong** (same specific sub-topic), **Related** (same chapter, different specific fact), **None** (not covered in your practice sets).

Subject	Exact	Strong	Related	None	Total	Coverage*
Physics	5	9	23	8	45	82.2%
Chemistry	4	11	28	2	45	95.6%
Biology	3	14	22	51	90	43.3%
TOTAL	12	34	73	61	180	66.1%

*Coverage = questions with an Exact/Strong/Related match, i.e. a similar concept exists somewhere in your practice sets.

Physics — Detailed Matches (Q1–45)

Q#	Set	Matched Topic / Question	Strength
1	NL	Motion of Bodies in Contact (Ch.11)	Related
2	NL	Pulley Systems (Ch.13)	Related
3	-	-	None
4	NL	Q45 - Atwood machine acceleration	Exact
5	NL	Newton's Second Law (Ch.5)	Related
6	-	-	None
7	NL	Pseudo Force (Ch.16)	Related
8	NL	Q63 - block in accelerating lift	Strong
9	NL	Spring Force (Ch.14, concept adjacent)	Related
10	-	-	None
11	NL	Pulley Systems (Ch.13)	Related
12	NL	Impulse (Ch.6)	Related
13	NL	Q13 - $F=ma$, velocity after time	Strong
14	NL	Impulse (Ch.6)	Related
15	NL	Q17 - variable force integration	Exact
16	NL	Impulse (Ch.6)	Related
17	NL	Normal Reaction / Pseudo Force	Related
18	NL	Q36 - apparent weight in lift	Strong
19	NL	Q70 - translational equilibrium	Strong
20	NL	Newton's Second Law (Ch.5)	Related
21	NL	Motion of Bodies in Contact (Ch.11)	Related
22	NL	Normal Reaction (Ch.10)	Related
23	-	-	None
24	-	-	None
25	NL	Pseudo Force (Ch.16)	Related
26	NL	Newton's Second Law (Ch.5)	Related
27	-	-	None
28	NL	Q68 - definition of translational equilibrium	Exact
29	NL	Impulse (Ch.6)	Related
30	NL	Q19/20 - cricket ball impulse examples	Strong
31	NL	Q66 - mechanical advantage of fixed pulley	Exact
32	-	-	None
33	NL	Impulse (Ch.6)	Related
34	NL	Motion of Bodies in Contact / Strings	Related
35	NL	Impulse (Ch.6, same concept)	Strong
36	NL	Pulley Systems / Strings	Related
37	NL	Q36 - same lift/apparent-weight formula	Strong
38	NL	Newton's Second Law (Ch.5)	Related

Q#	Set	Matched Topic / Question	Strength
39	-	-	None
40	NL	Pseudo Force (Ch.16)	Related
41	NL	Q23 - rocket thrust formula	Exact
42	NL	Normal Reaction (same apparent-weight formula)	Strong
43	NL	Impulse (Ch.6, calculus-based)	Related
44	NL	Q19 - impulse on rebounding ball	Strong
45	NL	Newton's Second Law (Ch.5)	Related

Chemistry — Detailed Matches (Q46–90)

Q#	Set	Matched Topic / Question	Strength
46	CB	Q16 - expanded octet concept	Strong
47	CB	Valence Bond Theory (Ch.5)	Related
48	CB	Q63 - NH ₄ ⁺ coordinate bond formation	Exact
49	CB	Q36 - SF ₄ /SF ₆ sp ³ d hybridisation	Strong
50	CB	Hybridisation (Ch.6)	Related
51	CB	VSEPR Theory (Ch.7)	Related
52	CB	Q35 - PCl ₅ sp ³ d, trigonal bipyramidal	Strong
53	CB	Q41 - sp hybridisation, linear, 180 degrees	Strong
54	CB	VSEPR Theory (Ch.7, Xe theme)	Related
55	CB	Q19 - formal charge calculation	Strong
56	CB	Q51 - why H ₂ O has smallest bond angle	Exact
57	CB	Covalent Bond (Ch.4)	Related
58	CB	Valence Bond Theory (Ch.5)	Related
59	-	-	None
60	CB	Bond Parameters (Ch.8)	Related
61	CB	Q56 - bond order vs bond enthalpy	Strong
62	CB	Covalent Bond (Ch.4, octet exceptions)	Related
63	CB	Q28 - sigma vs pi bond comparison	Strong
64	CB	Valence Bond Theory (Ch.5)	Related
65	CB	Q46 - VSEPR repulsion order	Exact
66	CB	Valence Bond Theory (Ch.5)	Related
67	CB	VSEPR Theory (Ch.7)	Related
68	CB	Hybridisation (Ch.6)	Related
69	CB	Valence Bond Theory (Ch.5)	Related
70	CB	VSEPR Theory (Ch.7, lone pair effect)	Related
71	CB	Hybridisation (Ch.6)	Related
72	CB	Q50 - SF ₄ see-saw shape (same logic)	Strong
73	CB	Q51 - H ₂ O vs NH ₃ vs CH ₄ angle comparison	Strong
74	CB	Hybridisation (Ch.6, sp ³ d ² general)	Related
75	CB	Valence Bond Theory (Ch.5)	Related
76	CB	Q41 - sp hybridisation concept	Related
77	CB	VSEPR Theory (Ch.7)	Related
78	CB	VSEPR Theory (Ch.7)	Related
79	CB	Q32 - pi bond restricts rotation	Exact
80	CB	Hybridisation (Ch.6)	Related
81	CB	Q40 - hybrid orbital incorrect-statement style	Strong
82	CB	Hybridisation (Ch.6)	Related
83	CB	VSEPR Theory (Ch.7, lone-pair-driven linear shape)	Related

Q#	Set	Matched Topic / Question	Strength
84	CB	Hybridisation (Ch.6, tetrahedral geometry)	Related
85	CB	Hybridisation (Ch.6)	Related
86	CB	Bond Parameters (Ch.8, bond order/length)	Related
87	CB	Covalent Bond (Ch.4, octet expansion limits)	Related
88	-	-	None
89	CB	Hybridisation (Ch.6)	Related
90	CB	Q48 - XeF4 square planar, lone pairs	Strong

Biology — Detailed Matches (Q91–180)

Q#	Set	Matched Topic / Question	Strength
91	-	-	None
92	AK2	Ch.1 - Five-kingdom classification criteria	Strong
93	-	-	None
94	AK2	Ch.1 - Classical vs modern taxonomy	Related
95	-	-	None
96	-	-	None
97	AK2	Ch.1 - Kingdom Monera (general)	Related
98	-	-	None
99	AK2	Ch.1 - Protista 'kingdom of convenience'	Exact
100	-	-	None
101	-	-	None
102	-	-	None
103	-	-	None
104	-	-	None
105	-	-	None (no source data)
106	-	-	None (no source data)
107	-	-	None
108	-	-	None
109	-	-	None
110	-	-	None
111	-	-	None
112	-	-	None
113	-	-	None
114	-	-	None
115	-	-	None
116	-	-	None
117	AK2	Ch.1 - Kingdom Monera (general)	Related
118	-	-	None
119	AK2	Ch.1 - style: Assertion-Reason 'both true, not explanation'	Related
120	AK2	Ch.1 - Protista as bridging kingdom	Strong
121	AK2	Ch.1 - Mycoplasma statement evaluation	Strong
122	-	-	None
123	-	-	None
124	-	-	None
125	AK2	Ch.1 - Cyanobacteria/heterocyst (general)	Related
126	-	-	None

Q#	Set	Matched Topic / Question	Strength
127	-	-	None
128	-	-	None
129	-	-	None
130	-	-	None
131	-	-	None
132	-	-	None
133	-	-	None
134	-	-	None
135	-	-	None
136	AK1	Ch.12 - Chordata notochord/vertebral column	Strong
137	AK2	Ch.15 - Pisces circulatory system (general)	Related
138	AK1/AK2	Echinodermata & Mollusca chapters (general)	Related
139	AK1	Ch.5 - Platyhelminthes (general)	Related
140	AK1	Ch.8 - Arthropoda open circulatory system	Strong
141	AK1/AK2	Coelenterata/Ctenophora + Platyhelminthes chapters	Related
142	AK1	Q55 - Reptilia true statement (oviparous etc.)	Strong
143	AK2	Ch.15 - Chondrichthyes lacks air bladder	Strong
144	AK1	Ch.4 - Coelenterata genus examples (general)	Related
145	-	-	None
146	AK1	Ch.16/19 - Amphibia & Mammalia chapters	Related
147	AK1	Q53 - Aves incorrect-statement style (teeth/structures)	Strong
148	AK1	Ch.1 - coelom/body-plan basics	Related
149	AK2	Q22 - ctenophore comb plates locomotion	Exact
150	AK2	Q51 - exclusively marine sets	Strong
151	AK2	Ch.15 - Pisces genus examples (general)	Related
152	-	-	None
153	-	-	None
154	AK1	Ch.7 - Nereis parapodia	Strong
155	AK1	Ch.1 - body cavity type matching	Strong
156	-	-	None
157	AK1	Ch.6 - Aschelminthes/roundworm features	Related
158	AK1/AK2	Porifera chapters (general)	Related
159	-	-	None (no source data)
160	AK1	Ch.4 - Obelia metagenesis (life cycle stages)	Related
161	-	-	None
162	AK2	Ch.13 - Protochordata notochord in larval tail	Strong
163	AK1	Ch.14 - Cyclostomata (general)	Related
164	-	-	None

Q#	Set	Matched Topic / Question	Strength
165	AK1	Ch.12 - post-anal tail as chordate feature	Strong
166	-	-	None (no source data)
167	-	-	None
168	-	-	None
169	AK1	Ch.8 - Arthropoda statements (general)	Related
170	-	-	None
171	-	-	None
172	-	-	None
173	AK2	Ch.9 - Mollusca foot/mantle (related concept)	Related
174	-	-	None
175	AK1/AK2	Annelida closed-circulation chapters	Related
176	-	-	None
177	AK1	Ch.16 - Amphibia statements (general)	Related
178	AK1	Q61 - Struthio/Ostrich Ratitae flightless	Exact
179	AK2	Ch.15 - Chondrichthyes cartilaginous (general)	Related
180	AK1	Q45 - Petromyzon dies after spawning (same genus)	Strong